Qwen

Certifying Cyboquatic Machinery: A Phased, Replay-Validated Path from Data Integrity to Production-Grade Ecosafety

Establishing the Data Trust Foundation with rcalib and rsigma

The advancement of the Cyboquatic ecosafety framework hinges on the foundational principle that no physical or ecological risk assessment can be trusted without a rigorous and mandatory gate for data quality. The strategic directive is unequivocal: complete and enforce the data-quality planes, `rcalib` (representing ingest schema fidelity) and `rsigma` (representing sensor uncertainty), before proceeding with the calibration of any new physical risk planes such as `rtox`, `rbiodiversity`, or `rcarbon` . This phased approach ensures that the system's behavior is anchored in verifiable data integrity, preventing the propagation of errors and uncertainties into more complex ecological models. The research objects within this category are not merely diagnostic tools but first-class citizens in the Lyapunov residual and Knowledge-Eco-Impact-Risk (KER) scoring, endowed with their own hard gates in deployment decisions . This section provides a comprehensive analysis of the mathematical kernels, integration logic, and governance structures for these two critical data trust planes, establishing the bedrock upon which all subsequent ecological risk assessments will be built.

The ingest-trust plane, `rcalib`, is defined by a scalar ingest-risk metric, `I`, which is a weighted sum of diagnostic counts observed over a closed time window . The minimal kernel proposed is a linear form where weights reflect the severity of different error classes . This scalar `I` is then normalized into a `RiskCoord` coordinate, `r_{\text{calib}}`, using a piecewise-linear corridor map with distinct safe, gold, and hard bands . This normalization process ensures that `r_{\text{calib}}` is a monotone increasing function of the underlying data quality issues; more severe or frequent ingest faults can only increase or maintain the value of `r_{\text{calib}}`, never decrease it . The full CEIM-form variant expands this kernel to include additional diagnostic counts such as missing schema rows, missing corridor rows, and unit inconsistencies, further enriching the signal of data trustworthiness .

The sensor-uncertainty plane, rsigma, complements rcalib by quantifying the uncertainty inherent in physical measurements. It is constructed as a composite coordinate derived from component risks: r_{\text{drift}}, r_{\text{noise}}, r_{\text{bias}}, and r_{\text{loss}} . Each component represents a distinct source of sensor error, such as long-term drift, short-term variance, systematic deviation from a lab standard, and data loss due to intermittent connectivity . Each of these components is first normalized into its own [0,1] coordinate using a dedicated corridor map with safe/gold/hard bands calibrated from sensor calibration histories . These individual risk coordinates are then aggregated into a single composite rsigma using a weighted Euclidean norm . This aggregation method preserves monotonicity, meaning an increase in any individual source of uncertainty (e.g., increased noise or bias) will cause rsigma to increase or remain flat, but never decrease . The entire structure of rcalib and rsigma is grounded in existing proofs of monotonicity and convexity, which guarantee that adding these terms to the Lyapunov residual will preserve its fundamental properties .

The integration of these data trust planes into the core framework is achieved through an extension of the Lyapunov residual, V_t. The original quadratic form, V_t = Σ(w_j * r_j^2), is augmented to create a "full" residual, V_t^{full} . This expanded residual explicitly includes non-negative quadratic terms for both rcalib and rsigma, weighted by their respective Lyapunov weights sourced directly from the ecosafety.corridors.v2 table . The formula for this extended residual is:

$$V_t^{\text{full}} = \sum_{j \in \text{physical}} w_j r_{j,t}^2 + w_{\text{calib}} r_{\text{calib},t}^2 + w_\sigma r_{\sigma,t}^2$$

This mathematical formulation has profound implications. Because all terms are non-negative, the addition of data quality planes ensures that the total risk metric V_t is a conservative measure; it can only increase or stay the same when data quality degrades, regardless of the state of the physical planes . The discrete safestep invariant, which requires V_{t+1} ≤ V_t outside a small interior region, continues to apply to this expanded state space, ensuring that the system trajectory remains stable and does not inadvertently move into a less-safe state due to poor data quality . Furthermore, rcalib and rsigma are integrated into the KER scoring. The risk-of-harm, R, is redefined as the maximum over all coordinates, including r_{\text{calib}} and r_{\sigma} . This means that any significant data quality issue will directly and immediately raise the R score, signaling a higher level of risk across the board . The governance layer is also strengthened by coupling these coordinates with trust scalars. A data trust scalar, D_{\\text{data}} = 1 - r_{\\text{calib}}, is introduced, which down-scales the credited knowledge (K_adj) and eco-impact (E_adj) scores . This creates a formal

mechanism where worse data quality can only reduce, not inflate, the perceived performance of the system, structurally preventing over-crediting under bad conditions .

To operationalize these concepts, several key research objectives must be addressed. First is the calibration and freezing of the ingest error kernel. This involves regressing the weight vectors `(w_miss, w_rfc, w_type, w_varid)` or `(m,s,c,u)` against labeled datasets of Phoenix ingest logs to ensure the weights accurately reflect the real-world impact of each error class . Concurrently, the corridor bands `(I_safe, I_gold, I_hard)` must be fit to these same logs, ensuring that "clean," "moderate problem," and "chronic failure" windows cluster in the desired low-to-high ranges of `r_{\text{calib}}` . Second is the completion of the `rsigma` path for Phoenix sensors. This requires defining the raw metrics for drift, noise, bias, and loss for each relevant sensor family (e.g., flow, pressure, turbidity) and calibrating their corresponding corridor bands based on calibration history and spec sheets . Third is the enforcement of these planes in every shard and deployment decision. The `rcalib` column must be marked as `mandatory=true` in all Phoenix shard families' schema definitions (`EcoNetSchemaShard2026v1`) . Furthermore, `kerdeployable` must be extended with explicit, independent gates like `BlockedByCalib`, which halts deployment if `r_{\text{calib}}` enters its hard band, irrespective of the status of any physical plane . Finally, a robust validation harness is needed. This involves creating replay jobs that inject synthetic ingest faults (e.g., dropping mandatory columns, mismatching headers) and sensor faults (e.g., injecting monotone drift, scaling noise) into historical Phoenix shards. These replays must assert that `r_{\text{calib}}`, `r_{\sigma}`, `V_t`, and `R` weakly increase, while `K_{adj}` and `E_{adj}` weakly decrease, providing a quantitative guarantee of the monotonicity properties .

Research Object	Description	Mathematical Kernel / Formula
Ingest Error Scalar (`I`)	Weighted sum of diagnostic counts (missing fields, RFC violations, type mismatches, etc.) indicating data ingestion health.	
Normalized Ingest Risk (`r_calib`)	Monotone `RiskCoord` in [0,1] derived from the ingest error scalar `I` via a corridor map.	r_{\text{calib}}(I) = \\begin{cases} 0 & I \le I_{\text{safe}} \\ \frac{I - I_{\text{safe}}}{I_{\text{gold}
Sensor Uncertainty Components	Normalized `RiskCoords` for individual sensor error modes: drift, noise, bias, loss. Each is mapped from a raw error metric via its own corridor.	for u ∈}, u_{gold}, u_{hard
Composite Sensor Uncertainty (`rsigma`)	Weighted L2 norm of component uncertainty coordinates, representing overall sensor reliability.	
Extended Lyapunov Residual (`V_t^{full}`)	Original residual plus non-negative quadratic terms for data trust planes.	
Trust-Adjusted KER (`K_adj`, `E_adj`)	Adjusted Knowledge and Eco-Impact scores	

Research Object	Description	Mathematical Kernel / Formula
	that are non-increasing functions of `rcalib`.	

By completing this work, the Cyboquatic framework establishes a certified baseline of data trust. The `rcalib` and `rsigma` planes become indispensable components of the ecosafety spine, transforming data quality from a background concern into a central pillar of safety governance. This foundational work enables all future extensions to be built upon a reliable and verifiable substrate, ensuring that subsequent physical risk calculations are meaningful and that the system's behavior remains predictable and safe even in the face of imperfect data.

Integrating Physical Risk Planes into the Unified Lyapunov Residual

With a robust data trust foundation established through the `rcalib` and `rsigma` planes, the next phase of advancing the Cyboquatic ecosafety framework involves the careful integration of new physical risk planes: `rtox` (toxicity), `rbiodiversity` (biodiversity impact), and `rcarbon` (carbon footprint). The guiding principle for this integration is to expand the scope of the existing `rx`–`Vt`–`KER` grammar rather than replacing it, thereby preserving the mathematical elegance and proven properties of the Lyapunov-based structure . The user's directive is clear: fold these new planes directly into the main `RiskVector` and the global Lyapunov residual `Vt`, treating them as first-class coordinates alongside energy, hydraulics, and data quality . This approach maintains a single, holistic measure of system safety while allowing for nuanced trade-off analysis between different types of risk. The mathematical justification for this expansion rests on the well-established property that adding a non-negative quadratic term to a convex function preserves its convexity and non-negativity, thus ensuring the discrete Lyapunov argument—that the system trajectory remains non-expansive in risk—continues to hold .

Each new physical plane is defined by a set of research objects that follow a consistent pattern. At its core is a physics kernel that maps raw physical or environmental measurements into a normalized `RiskCoord` in the `[0,1]` interval. For the carbon plane, `rcarbon`, this kernel is based on the Carbon-Enhanced Impoundment Model (CEIM), calculating net CO_2e mass load from influent and effluent concentrations and

flow rates, and normalizing it by the energy consumed . The resulting intensity metric (kg CO_2e/kWh) is then mapped into `r_{\text{carbon}}` using a corridor map with strongly negative "safe" bands (for carbon-negative operation), near-neutral "gold" bands, and positive "hard" bands . The goal is to design the kernel so that any mode of net-positive carbon emissions pushes `r_{\text{carbon}}` towards 1, thereby raising `Vt` and making the step a candidate for rejection by `safestep` .

For the materials plane, which contributes to `rtox`, the kernel involves modeling material kinetics, such as first-order decay to determine a `t90` (time to 90% mass loss) . This `t90` value is normalized into a `r_{t90}` coordinate, with slower decay (longer `t90`) mapping to higher risk . Beyond decomposition, the toxicity plane aggregates multiple sub-risks—including ecotoxicity, micro-residue potential, leachate CEC risk, and PFAS-like residual risk—each normalized into its own coordinate . These sub-coordinates are then combined using a convex aggregation method, typically a weighted quadratic form, to produce a composite `r_{\text{materials}}` or `r_{\text{tox}}` . This ensures that an increase in any sub-risk, such as higher ecotoxicity or longer `t90`, will cause the composite risk to increase or remain constant, never decrease . The final step is to freeze these plane-specific kernels and their parameters in a "Plane v1" specification document, which formally defines the physics, normalization, and contribution to `Vt`, along with associated unit tests that serve as executable proofs of monotonicity and boundedness .

Similarly, the biodiversity plane, `rbiodiversity`, is operationalized by mapping three key habitat metrics—connectivity index, structural complexity, and colonization score—from hydrodynamic and GIS models into their respective `RiskCoord` values . Unlike typical risk metrics, these are designed as "inverse-good" corridors, where higher values of the physical metric (e.g., better habitat connectivity) correspond to lower risk values (closer to 0) . An increase in any of these physical metrics lowers `r_{\text{biodiversity}}`, while a degradation in habitat quality raises it. These normalized sub-coordinates are aggregated into a composite `r_{\text{biodiversity}}` using a weighted quadratic kernel, which guarantees that any drop in connectivity or complexity will monotonically increase the overall `rbiodiversity` coordinate and, consequently, the Lyapunov residual `Vt` .

The following table summarizes the core research objects and mathematical kernels for each of the new physical planes, illustrating the consistency and rigor applied to their integration.

Plane	Core Physics Kernel	Normalization & Aggregation	Contribution to `Vt`
Carbon (`rcarbon`)	Net Intensity: $I = (-M_{sequestered}/E) + I_{grid}$, where M is mass load and E is energy .	Piecewise-linear corridor map: $r_{\text{carbon}} = \text{corridor}(I; I_{safe}, I_{gold}, I_{hard})$. Strongly negative `safe` band .	$w_{\text{carbon}} \cdot r_{\text{carbon}}^2$
Materials/Toxicity (`rtox`)	Sub-coordinates: r_{t90} (from $t_{90} = \ln(10)/k$), r_{tox}, r_{micro}, r_{leach}, $r_{PFASresid}$.	Convex aggregation: $r_{\text{materials}} = \sqrt{\sum_i w_i r_i^2}$, ensuring monotonicity in each sub-risk .	$w_{\text{materials}} \cdot r_{\text{materials}}^2$
Biodiversity (`rbiodiversity`)	Sub-coordinates: r_{conn} (from Connectivity Index), r_{comp} (from Structural Complexity), r_{colon} (from Colonization Score) .	Inverse-good corridor maps (higher good → lower risk) followed by quadratic aggregation: $r_{\text{bio}} = \sqrt{w_c r_c^2 + w_s r_s^2 + w_o r_o^2}$.	$w_{\text{biodiversity}} \cdot r_{\text{biodiversity}}^2$

The integration of these planes into the `RiskVector` and `Vt` is not just a matter of adding new fields; it is a governance task that requires unifying everything under common ALN schemas and enforcing a "no corridor, no build" rule . The `ecosafety.corridors.v2` schema must be updated with mandatory rows for all new varids (e.g., `CARBON.NETINTENSITY`, `BIODIV.CONNECTIVITY`, `MATERIAL.T90DAYS`), each specifying safe/gold/hard bands and a Lyapunov weight . Similarly, the `ecosafety.riskvector.v2` schema must be updated to include the new plane coordinates . Any attempt to reference a varid in code or a shard without a corresponding row in `ecosafety.corridors.v2` must fail CI, ensuring that all risk coordinates are traceable to a formal definition . This governance layer, combined with the formal proofs of monotonicity and Lyapunov preservation, transforms the framework from a collection of isolated risk calculators into a certifiable, production-grade system where every accepted change must tighten eco-risk envelopes or maintain them while increasing traceability . By freezing the kernels and corridors for these new planes *after* the data trust foundations are live, the framework ensures that corridor calibration runs in a world where the impact of data quality itself is already being measured and managed, leading to more robust and realistic risk assessments .

Implementing Non-Compensability Constraints and Dual Safestep Invariants

While integrating the new physical risk planes into a unified Lyapunov residual provides a powerful, holistic measure of system safety, it introduces a potential vulnerability: the possibility of harmful trade-offs. An improvement in one plane's risk coordinate could theoretically compensate for a worsening in another, potentially masking unacceptable levels of harm. To address this, the Cyboquatic framework must implement explicit,

enforceable governance rules that codify non-compensability. The user's directive specifies the creation of "explicit, ALN-encoded dual safestep constraints that prevent harm in critical planes... from being offset by improvements in others" . This governance layer acts as a crucial ethical and ecological safeguard, moving beyond the purely mathematical optimization of the global Lyapunov residual to incorporate domain-specific principles that certain forms of harm are non-negotiable.

The implementation of this concept relies on a governance layer encoded in Application Logic Notation (ALN), which allows for the expression of high-level, machine-checkable rules about acceptable system behavior. These rules manifest as dual safestep constraints or cross-plane invariants. The core idea is to define specific planes—namely `rtox`, `rbiodiversity`, and potentially `rcarbon` and `rmaterials`—as having "non-offsettable" status . For these planes, an increase in risk above a specified threshold (e.g., the "gold" band) must have a direct and immediate consequence on the deployment decision, irrespective of concurrent improvements in other planes. This is operationalized through ALN predicates parameterized by the affected plane. For instance, a predicate named `NoOffsetAboveGold(p)` would be defined for a critical plane `p` . This invariant would stipulate that if the risk coordinate `r_p` for plane `p` were to cross its gold threshold upward during a transition, then the overall system risk `R` must not decrease and the eco-impact `E` must not increase . Formally, for a transition at timestep `t+1`, if `r_{p,t} < r_{p,gold} ≤ r_{p,t+1}`, then the invariant requires that `R_{t+1} ≥ R_t` and `E_{t+1} ≤ E_t` . Such a constraint would be enforced as part of the `kerdeployable` check; a transition that violates this rule would be rejected.

These non-compensability invariants are a refinement of the global `safestep` rule. While `safestep` governs the global Lyapunov residual `Vt`, these new ALN invariants govern specific, high-stakes coordinates within that residual. They provide a way to enforce qualitative, domain-driven safety requirements that might not be fully captured by the quantitative sum of squares in `Vt`. For example, a marginal increase in `rcarbon` that slightly raises `Vt` might be tolerable if it leads to a large improvement in `rhydraulics`. However, a similar marginal increase in `rbiodiversity` that crosses the gold line should be treated as a serious event, as it represents a tangible, irreversible loss of ecosystem health. The `NoOffsetAboveGold(biodiv)` invariant ensures that this event is flagged and cannot be papered over by gains elsewhere . This distinction is critical for aligning the technical system with the overarching mission of "help just by researching it," ensuring that every accepted change genuinely improves or at least maintains the ecological balance .

A related concept is the use of a dual residual and a corresponding `safestep_dual` contract. This approach involves partitioning the risk coordinates into two categories:

those contributing to the main physical risk residual V_t (e.g., r_{energy}, $r_{biodiversity}$, r_{carbon}) and those contributing to a separate uncertainty or data-quality residual U_t (e.g., r_{calib}, r_{sigma}) . A dual safestep contract would then require that *both* residuals be non-increasing outside their respective interiors: $V_{t+1} \leq V_t$ AND $U_{t+1} \leq U_t$. This provides a clean separation of concerns, allowing for distinct management strategies for physical risk and data trustworthiness. For instance, the governance thresholds for V_t (the PROD lane gates of $K \geq 0.90$, $E \geq 0.90$, $R \leq 0.13$) could apply independently of the thresholds for U_t, which might be used to manage sensor maintenance schedules or data pipeline health . While the user's primary recommendation is to keep everything in one residual for simplicity, the dual residual concept provides a valuable alternative or complementary structure for managing different facets of risk . The proofs supporting this structure show that adding extra non-negative quadratic terms to the residual preserves the Lyapunov properties, and encoding this with a parallel U_t residual matches the described dual-residual design .

The implementation of these advanced governance mechanisms requires several concrete steps. First, the non-compensability invariants must be formally defined and encoded in ALN. This involves writing out the logical conditions for NoOffsetAboveGold(carbon), NoOffsetAboveGold(biodiv), etc., and binding them to the specific Lyapunov channels ("carbon", "biodiversity") they govern . These ALN invariants must then be wired into the kerdeployable configuration, so they are actively checked during deployment decisions, not just logged as warnings . Second, the CI/CD pipeline must be enhanced to validate these invariants. This requires developing targeted replay scenarios that specifically test for violations. For example, a replay harness would need to generate artificial transitions where rbiodiversity increases above its gold line while simultaneously improving other plane coordinates. The replay job would then assert that such a scenario fails the NoOffsetAboveGold(biodiv) check, thereby catching regressions in the invariant logic . Third, per-plane KER views should be considered. Defining separate KER metrics for each plane (K_{plane}, E_{plane}, R_{plane}) could provide finer-grained insights into what is limiting deployability and make it easier to enforce these non-compensability rules . Finally, long-horizon governance metrics should be tracked. This involves creating governance shards that monitor the running medians of critical plane coordinates like r_{carbon} and $r_{biodiversity}$ over multi-year periods, with explicit targets for eco-restoration (e.g., $\lim_{T\to\infty} \text{median}_{t\leq T}\, r_{carbon}(t) = 0$) . Any change to the system that would cause these long-term medians to worsen would be heavily scrutinized, reinforcing the principle that short-term optimizations cannot come at the expense of long-term ecological goals. By layering these explicit, enforceable rules on top of the unified Lyapunov residual, the Cyboquatic framework achieves a much higher degree of safety assurance and alignment with its core ecological principles.

Validation and Enforcement via Phoenix-Anchored Replay CI

The theoretical architecture of the Cyboquatic ecosafety framework—spanning phased governance, unified residuals, and explicit non-compensability constraints—is rendered meaningless without a rigorous and continuous validation mechanism. The provided materials establish Phoenix-anchored replay CI as the cornerstone of this validation strategy, serving as the primary tool to translate abstract mathematical properties into concrete, evidence-based guarantees of system behavior . This is not a static, one-time proof but a dynamic, automated feedback loop woven into the development lifecycle. Every change to a plane kernel, corridor band, or governance invariant is subjected to quantitative stress-testing against a corpus of real-world data from the Phoenix pilots (Lake Pleasant, Gila, MAR vaults, biodegradable substrates) . This process ensures that the framework's behavior remains predictable, safe, and aligned with the core research goal of tightening eco-risk envelopes.

The Phoenix replay CI harness operates through several distinct but interconnected testing paradigms. The first and most fundamental is **fault injection**. This paradigm is used to verify the monotonicity properties of the data trust planes, `rcalib` and `rsigma`. Replaying historical Phoenix shards through the current version of the framework serves as a trusted baseline. Then, synthetic faults are injected into the data streams of these replayed shards, and the results are compared. For `rcalib`, this involves injecting ingest-related faults such as dropping mandatory columns, introducing header mismatches between schema versions, or adding unknown varids . The assertion is that each of these faults must cause the calculated `r_{\text{calib}}` to increase or remain flat, which in turn must cause the Lyapunov residual `Vt` and risk-of-harm `R` to weakly increase, while the trust-adjusted K/E scores weakly decrease . For `rsigma`, the fault injection focuses on sensor data. This includes injecting monotone drift into a time series, scaling up the noise variance, or creating intermittent data dropouts . Again, the expected outcome is a demonstrable increase in `r_{\sigma}`, leading to a corresponding increase in `Vt` and `R` and a decrease in `K_{adj}` and `E_{adj}`. Any regression in this behavior, where a fault causes a *decrease* in risk, constitutes a CI failure and blocks the merge, providing a direct, quantitative guarantee that the system correctly penalizes data quality degradation .

The second major paradigm is **corridor evolution testing**. As the framework matures, the corridor bands for various risk planes will be tightened to reflect improved understanding or stricter regulatory standards. Phoenix replay is the essential tool for validating that these changes have the intended effect. When a corridor is changed—for

example, the "hard" band for `r_{t90}` is moved to a shorter time to enforce faster decomposition—the same set of historical Phoenix shards is replayed under both the old and new corridor definitions . The results are then compared using a "kerupgradeok" invariant. This invariant asserts that for any given trajectory, the new, tighter corridors must produce KER scores that are no better than the old ones: `K_new ≤ K_old`, `E_new ≤ E_old`, and `R_new ≥ R_old` . This ensures that upgrading the framework's specifications cannot retroactively make past behavior appear safer or more effective. It prevents a dangerous situation where a seemingly benign change to a corridor definition could mask a previously detected problem. This replay-based validation is critical for maintaining the integrity of the KER scoring over time .

The third paradigm is the direct **verification of non-compensability invariants**. The ALN-encoded dual safestep constraints, such as `NoOffsetAboveGold`, must also be continuously tested. The Phoenix replay harness is adapted to run targeted scenarios designed to test these specific rules. For example, to test `NoOffsetAboveGold(biodiv)`, the replay system would identify transitions in the historical Phoenix data where `rbiodiversity` naturally crossed its gold threshold. It would then check whether, in those transitions, the overall `R` score decreased or the `E` score increased, which would constitute a violation of the invariant. The harness would also be used to test for counterfactual scenarios: it would artificially force `rbiodiversity` to increase while other planes improve, and assert that the `NoOffsetAboveGold(biodiv)` check fails, thereby catching any bugs in the invariant's implementation . This active testing of the governance rules ensures that the ethical guardrails are not just theoretical but are robustly enforced in practice.

To make this entire validation suite manageable and generalizable, the research materials suggest packaging the components into explicit, reusable artifacts. This involves creating **Phoenix replay profiles**, such as "Phoenix-MAR-ecosafety" and "Phoenix-substrate-kinetics," which instantiate the generic replay harness with Phoenix-specific corridor tables, schema shards, and shard families . It also involves developing concrete **fault libraries**—versioned Rust crates containing pure functions like `inject_drift` or `inject_missing_columns`—that can be parameterized and reused across different profiles and basins . Finally, the **replay invariants** themselves are formalized in ALN specs, turning informal assertions into machine-checkable admissibility conditions for a successful CI run . This structured approach transforms the Phoenix replay from a collection of ad-hoc scripts into a scalable, certifiable validation engine. By anchoring the entire framework to this rigorous, evidence-based feedback loop, the Cyboquatic project ensures that its governance model is not just mathematically sound on paper, but practically robust and safe in the field.

Actionable Roadmap for Framework Advancement

Advancing the Cyboquatic ecosafety framework to a production-grade state requires a disciplined, multi-phase roadmap that systematically builds upon verified foundations. The synthesis of the provided materials yields a clear, actionable plan divided into three distinct stages: (1) establishing the data trust foundation, (2) expanding physical risk plane integration, and (3) enforcing non-compensability and long-term governance. This structured approach, grounded in the principles of Lyapunov stability and validated by Phoenix-anchored replay CI, ensures that every step tightens the framework's safety posture and aligns with the core objective of making every accepted change genuinely improve or maintain ecological outcomes.

Phase 1: Establish the Data Trust Foundation The primary objective of this phase is to finalize, calibrate, and enforce the `rcalib` (ingest schema fidelity) and `rsigma` (sensor uncertainty) planes, making them mandatory cross-cutting concerns before any new physical risk planes are introduced.

- **Key Actions:**

```
1.  <strong>Calibrate the Ingest Error Kernel:</strong> Using real-world P
2.  <strong>Complete the rsigma Aggregation Pipeline:</strong> For each ke
3.  <strong>Integrate and Enforce in Governance:</strong> Update the `Risk
4.  <strong>Implement Phoenix Replay Validation:</strong> Develop and wire
```

Phase 2: Expand Physical Risk Plane Integration Once the data trust foundation is certified as live and enforcing, the focus shifts to freezing and tightening the corridor bands for the new physical risk planes: `rtox`, `rbiodiversity`, and `rcarbon`.

- **Key Actions:**

```
1.  <strong>Finalize Plane Kernels and Corridors:</strong> For each of `rt
2.  <strong>Document and Integrate into Spine:</strong> Create detailed "P
3.  <strong>Extend Phoenix Replay for Corridor Evolution:</strong> Augment
```

Phase 3: Enforce Non-Compensability and Long-Term Governance The final phase solidifies the framework's safety guarantees by codifying cross-plane trade-off rules and establishing mechanisms for long-term ecological stewardship.

- **Key Actions:**

```
1.  <strong>Formalize Non-Compensability Invariants:</strong> Define and e
2.  <strong>Implement Dual Safestep and Per-Plane Views:</strong> Explore
3.  <strong>Establish Long-Horizon Governance Metrics:</strong> Create gov
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By meticulously following this three-phase roadmap, the Cyboquatic framework evolves from a conceptual model into a certifiable, production-grade system. It systematically builds a hierarchy of trust, starting from the ground level of data quality and progressively incorporating higher-level ecological risks, all while being continuously validated against real-world data. This disciplined, evidence-driven approach ensures that every contribution to the framework genuinely advances its mission of creating industrial machinery that is energy-efficient, carbon-negative, and ecologically restorative.

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